ii. Pathogenecity

The effect of microbes that cause diseases in plants. Example: *Xanthomonas citri*

2. Abiotic Stresses

Abiotic stress may occur due to an atmospheric condition (atmospheric stress) or soil condition (edaphic stress). Atmospheric stresses may occur due to excess and deficient levels of light, temperature and air pollutants.

i. Light Stress

Light limits the distribution of species. In low light intensity **Sciophytes** (shade loving plants) develop, while in high light intensity **Heliophytes** (high light loving plants) develop. In low light intensity, stomata do not fully open hence there is less diffusion of gases. As a result, there is less photosynthesis and the chlorophyll synthesis is also affected. High light intensity also inhibits photosynthesis. Change in photoperiod inhibits flowering.

ii. Temperature

Plants are adapted to a particular region and they face temperature stress in another region.

a. High temperature

High temperature causes soil and atmospheric drought. Plants are subjected to permanent wilting in soil drought and temporary wilting in atmospheric drought. Plants generally die above temperature of 44°C. However, some organisms like *Mastigocladus* (a cyanobacterium) grow well at 85°C to 90°C in hot springs. At 42°C synthesis of normal protein declines and new protein called Heat Shock Proteins (HSPs) appears. These proteins were discovered in fruit fly (*Drosophila melanogaster*) and since then they have also been discovered in animals, plants and microorganisms. At high temperature all physiological processes decline. Photosynthesis decreases and respiration increases. So, plants face a shortage of organic substances.



Apple, a temperate plant, when planted in tropical condition fails to produce fruits and growth is also affected.

b. Low Temperature

Low temperature stress is quite harmful to plants and the temperature near freezing point causes irreversible damage so that the plants fail to survive under extreme cold conditions. However, some plants growing in alpine and arctic regions can survive under low temperature and such plants are said to be **cold resistant**. Stress due to freezing temperature is called **frost stress**. Temperature below 10°C, decreases root growth, increases leakage of ions and ethylene production.



Some plant parts like Seeds, pollen grains and embryos can be stored at very low temperature (–196°C).

iii. Air pollutants

Important atmospheric pollutants prevalent in the Indian sub-continent are CO_2, CO, SO_2, NO_2, O_3, fluoride and H_2S. These pollutants do not cause visible injury but cause hidden injury. If the concentration of these pollutants increases visible injury like chlorotic and necrotic spots appear on leaves as well as inhibit photosynthetic carbon metabolism and biomass formation. Some pollutants at low concentration stimulate plant growth. Example: SO_2, NO_2 and NO. Respiration and photorespiration are sensitive to air pollutants. If the concentration of air pollutants is high, it inhibits respiration whereas at lower concentration stimulates respiration. Nitrogenous air pollutants under chronic exposure increases chlorophyll content while NO_2 reduces pigment content at acute exposure.

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